

DIEGO LOPEZ MAGDALENO

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EDUCATION

CETYS University

B. Sc. in Software Engineering

Ensenada, B.C. Graduating July 2027
GPA 3.8/4

WORK EXPERIENCE

Software Engineering Intern - Meta

Menlo Park, June 2026 - August 2026

Full-stack features for Instagram's logged-out discovery surfaces (instagram.com/popular)

- Shipped **12 production features** across Instagram's logged-out web surfaces serving **3.1M+ daily visitors** and **199M+ page views** — designed and ran A/B experiments (**150M+ exposures**) validating each feature's impact before full rollout across React/Relay, Hack/GraphQL, and Dataswarm.
- Built a **full-stack Category Hub (/popular/{broad-topic})** system end-to-end (static config → topic fetchers → GraphQL resolvers → React components → Instagram Web routing → controller-level gating), launching **54 SEO-optimized** pages serving **2.7M daily visits** with **91M+ organic search-driven sessions**.
- Designed an **AI content curation pipeline** using LLaMA 3.3 70B: Seed categories into Hive → Dataswarm AsyncOperator triggers per-category inference → model selects top-5 trending topics → results cached in TAO Ent for real-time serving with heuristic fallback.
- Re-architected the Galileo (/popular) discovery home with scroll-based lazy pagination, responsive dynamic grids, and **immersive reel** modals, **achieving a +6.6% traffic uplift (390K daily visits)** across **13.6M unique sessions**.

Software Engineering Intern - Meta

Menlo Park, June 2025 - August 2025

ClinicAll Project & Metrics Core Capabilities Team Contributions.

- Designed and developed ClinicAll, a hospital management system with role-based authentication and authorization using **Auth0, React, Node.js, PostgreSQL, and Prisma**.
- Built a real-time notification system for patient appointments using **Socket.IO**, with notifications stored in a **priority queue** and processed via scheduled cron jobs.
- Implemented a drug interaction engine to detect conflicts between drugs and symptoms, modeling complex many-to-many relationships in a **graph database structure** for efficient querying.
- Integrated custom typeahead search for drugs (FDA API) and symptoms, **improving search speed and accuracy**; added **AI-powered** interaction checks.
- Contributed to the Metrics Core Capabilities team, delivering bug fixes and improvements within **Meta's internal codebase**, while collaborating in discussions to ensure reliability of metrics systems.

MAJOR PROJECTS

AURA Platform Remote - Semptra Infrastructure National Hackathon 1st Place

Ensenada, B.C. January 2026 - May 2026

Industrial Risk Training Simulation Engine & LMS Contributions.

- Designed and developed the AURA platform, a specialized **learning management system** for industrial risk training with secure role-based navigation and authentication using **React, TypeScript, and Supabase**.
- Built an interactive, full-screen **simulation engine** for risk assessment training, **successfully tracking and evaluating user performance** through responsively placed **hotspots and conduct-based scoring**.
- Architected a customized **AI collaboration workflow** by engineering a strict file-based context system, enabling the AI to autonomously adhere to project specifications, track historical decisions, and **execute tasks with high contextual accuracy**.

Clash Royale Emote Recognition

Ensenada, B.C. January 2026 – February 2026

Computer vision system for recognizing emotes from facial expressions and hand gestures.

- Built a **computer vision pipeline** to classify in-game emotes from Clash Royale using **MediaPipe landmark extraction** (hands + facial blendshapes), engineering a 135-feature dataset from gesture and facial expression signals.
- Designed and trained **Random Forest and XGBoost** classifiers with feature selection and scikit-learn pipelines, achieving **~99.99% ROC-AUC** on multi-class emote classification.
- Developed a reproducible **ML inference pipeline in Python** (NumPy, Pandas, scikit-learn) and exported the trained model with Joblib for real-time prediction using CV2.

LEADERSHIP & EXTRACURRICULAR

InternClub

Ensenada, B.C. Summer 2023 -

Co-leader of a community of college students focused on developing programming skills and building real-world experience to land internships

- Mentored 20+ students** in resume building, project creation and interview preparation resulting in **3 internship placements**.
- Collaborative environment:** Work in teams on challenging projects to simulate real-world coding tasks.
- Hands-on coding practice:** Build real-world projects and solve problems to sharpen your skills.

SKILLS

PROGRAMMING LANGUAGES

Proficient: Javascript, Python, Java
Familiar With: C, C++

TECHNOLOGIES

Pandas, Scikit-Learn, Matplotlib, Numpy, React.js, Redux.js, node.js, Express, MySQL, MongoDB (NoSQL), Firebase, Git, SpringBoot, HTML, CSS, Tailwind, Supabase, JSON, IntelliJ, VSCode, WebSockets, Figma and Vercel.

AWARDS

Semptra Infrastructure Innovation Hackathon 1st place (2026)

NASA Space Apps Hackathon Participation (2024)

3rd Place and multiple participations at the State Math Olympiad (2022)